ACD1200 Datasheet

Infrared Carbon Dioxide Sensor

- Excellent long-term stability
- Good selectivity
- No oxygen dependence
- Standard digital output
- Quick response
- Fast recovery time
- Strong anti-interference ability

Product Summary

ACD1200 uses the non-dispersive infrared (NDIR) principle to detect the concentration of carbon dioxide in the air. ACD1200 has the advantages of good selectivity, no oxygen dependence, long life, etc. It has a digital interface output for easy use. ACD1200 is a high-performance sensor that integrates infrared absorption gas detection technology, precision optical path and high-precision signal detection circuit.

Applications

ACD1200 has a wide range of application scenarios and is suitable for air quality monitoring equipment, fresh air systems, air purification equipment, HVAC equipment, etc.

Figure 1.ACD1200 photo

1. Working principle

The ACD1200 sensor is designed based on the spectral absorption principle of CO2 gas. As shown in Figure 2, the sensor consists of three parts: a light source, an air chamber, and an infrared receiver. The infrared receiver is used to measure the intensity of infrared light emitted by the light source. When CO2 flows into the air chamber through the gas inlet and flows out of the air chamber from the outlet, CO2 absorbs infrared light of a specific wavelength in the air chamber, and changes in CO2 concentration will affect the amount of infrared absorption. When the concentration of CO2 changes, the intensity of the light source signal received by the infrared receiver will also change. The receiver calculates the CO2 concentration by detecting the absorption rate of infrared light of a specific wavelength.

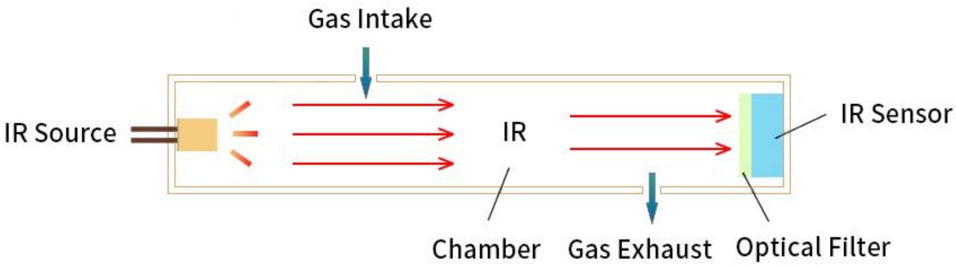

Figure 2. Working Principle of ACD1200

2. Technical Parameters

Table 1. Technical Parameters

NO.	Parameters	Description
1	Work Voltage	4.75 ~ 5.25V
2	Work Current	Average current, <45mA
3	Typical Power	100mW @5V
4	Measuring Range	400 ~ 5000ppm
5	Accuracy	±（50ppm+5% reading)
6	Preheating Time	120s
7	Operation Requirement	0℃ ~ 50℃； 0 ~ 95%RH (non-condensation)
8	Condition of Storage	-20℃ ~ 60℃； 0 ~ 95%RH (non-condensation)
9	Data Refresh Frequency	2s
10	Life Span	> 10year
11	Data	I^2C/UART

User Guide

1. Calibration Function Description

The sensor has automatic calibration and manual calibration functions, which can be selected by sending commands through the host (see 2.3.2 and 2.4.3 command lists). The sensor is set to automatic calibration mode by default when it leaves the factory.

1.1 Automatic Calibration Function

The sensor has a built-in automatic calibration algorithm, which can automatically calibrate and correct measurement errors periodically. The sensor completes an automatic calibration 24 hours after power-on, and then completes an automatic calibration every 7 days (168 hours). To ensure the accuracy after calibration, the CO2 concentration in the working environment of the sensor should be close to the outdoor atmospheric level for at least 1 hour within 24 hours after power-on and within 7 days of continuous operation.

1.2 Manual Calibration Function

Place the sensor in a working environment with a known CO2 concentration for more than 20 minutes, and send a manual calibration command through the host (see 2.3.2 and 2.4.3 command lists).

2. Interface Definition and Communication Protocol

2.1 ACD1200 Pin Assignment

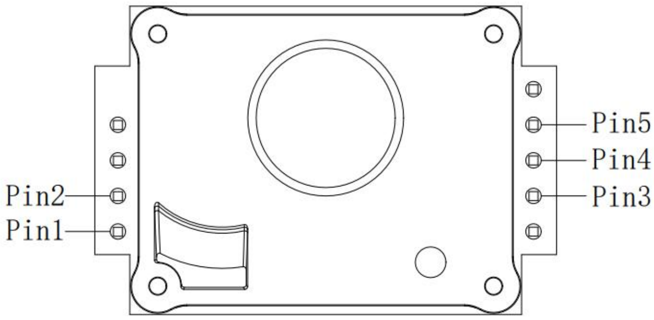

Figure 3.ACD1200 pin

Table 2. Pin definition

NO.	Pin	Name	Function
1	Pin 1	VCC	Power
2	Pin 2	GND	GND
3	Pin 3	SDA/RX	(sensor)Receive
4	Pin 4	SCL/TX	(sensor)Send
5	Pin 5	SET	Set communication method

2.2 Communication interface

The sensor provides I^2C and UART communication interfaces, which users can choose according to their needs. The input level of the interface is compatible with 3V to 5V, and the output level is internally pulled up to 5V. The sensor defaults to I^2C communication (Pin 5 is left floating, built-in 4.7 pull-up resistor); when Pin 5 is connected to low level, it is UART serial communication.

2.3 I^2C communication protocol

When the sensor uses I^2C communication mode, it communicates according to the standard I^2C protocol, and the communication rate supports up to 100kHz.

2.3.1 CRC check method

Table 3.CRC check

NO.	Project	Value
1	Name	CRC-8
2	Multi-nomial	0x31 $(x^8+x^5+x^4+1)$
3	Initial Value	0xFF
4	Example	CRC(0x0000)=0x81

2.3.2 Command list

Table 4. I^2C communication list

NO.	Function	Command
1	read CO_2 concentration	0x0300
2	set/ read calibration method	0x5306
3	set/ read manual calibration value	0x5204
4	Set / Read the restore factory values	0x5202
5	read software version	0xD100
6	read sensor SN	0xD201

1) Read CO2 concentration command, including downlink instructions and uplink data, as shown in Table 5 and Table 6.

Table 5. Downlink Instruction Sheet

Start bit	Address	Command high bytes	Command low bytes	Stop bit
Start	0x54	0x03	0x00	Stop

Table 6. Uplink Data Sheet

Start byte	Add	The highest concentration byte	Concentration sub-high byte	CRC (first 2 bytes)	Concentration sub-low byte	Min concentration byte	CRC (first 2 bytes)	High temp bytes	Temp low bytes	CRC (first 2 bytes)	Stop byte
Start	0x55	PPM3	PPM2	CRC1	PPM1	PPM0	CRC2	TEMP 1	TEMP 2	CRC3	Stop

2) Set the calibration mode command to the manual or automatic mode, as shown in Table 7.

Table 7. Calibration Mode Instruction Sheet

Start bit	Address	Command high bytes	Command low bytes	Data high bytes	Data low bytes	CRC (first 2 bytes)	Stop bit
Start	0x54	0x53	0x06	0x00	0x00	0x81	Stop

Note: When the low byte of the data is 0, it is manual calibration, and when it is 1, it is automatic calibration.

Read the calibration mode command, as shown in Table 8 and Table 9, which includes downlink instructions and uplink responses, and is used to determine whether the calibration mode setting command is written correctly. When the response data in Table 9 is the same as the write data in Table 7, it indicates that the calibration mode setting command is written successfully. The instructions in Tables 8 and 9 do not need to be used in pairs with the calibration mode setting command in Table 7; after the calibration mode setting command in Table 7 is sent, the interval time must be greater than 5ms before the instructions in Tables 8 and 9 can be used.

Table 8. Downstream command list

Start bit	Address	Command high bytes	Command low bytes	Stop bit
Start	0x54	0x53	0x06	Stop

Table 9. Upside Response Table

Start bit	Address	Data high bytes	Data low bytes	CRC (first 2 bytes)	Stop bit
Start	0x55	0x00	0x00	0x81	Stop

Note: When the low byte of the data is 0, it is manual calibration, and when it is 1, it is automatic calibration.

3) Manual calibration command (single point calibration), set the reference point to the user-specified value, as shown in Table 10.

Table 10. Manual calibration instruction table

Start bit	Address	Command high bytes	Command low bytes	Data high bytes	Data low bytes	CRC (first 2 bytes)	Stop bit
Start	0x54	0x52	0x04	0x01	0xC2	0x50	Stop

Note: The data is the calibration target value, the unit is ppm, the example is 450ppm.

The command to read the manual calibration value, as shown in Table 11 and Table 12, includes downlink instructions and uplink responses, which are used to determine whether the value written by the manual calibration command is correct. When the response data in Table 12 is the same as the write data in Table 10, it means that the manual calibration command is written successfully. The instructions in Tables 11 and 12 must be used in pairs with the manual calibration command in Table 10, and no other instructions can be inserted in between; after the manual calibration command in Table 10 is sent, the interval time must be greater than 5ms before the instructions in Tables 11 and 12 can be used.

Table 11. Downstream command list

Start bit	Address	Command high bytes	Command low bytes	Stop bit
Start	0x54	0x52	0x04	Stop

Table 12. Up-line Response Table

Start bit	Address	Data high bytes	Data low bytes	CRC (first 2 bytes)	Stop bit
Start	0x55	0x01	0xC2	0x50	Stop

Note: The data is the read calibration target value in ppm, the example is 450ppm.

4) Restore factory settings command. When necessary, use this command to restore to factory values, as shown in Table 13.

Table 13. Restore factory settings command list

Start bit	Address	Command high bytes	Command low bytes	Data	Stop bit
Start	0x54	0x52	0x02	0x00	Stop

Read the operation results of restoring factory settings, as shown in Table 14 and Table 15, including downstream instructions and uplink responses, which are used to determine whether the restoration of factory settings is successful. When the response data in Table 15 is 1, it means that the factory settings are restored successfully. After the factory reset command in Table 13 is successfully written, the uplink response data can be read multiple times continuously.

Table 14. Downlink Instruction Sheet

Start bit	Address	Command high bytes	Command low bytes	Stop bit
Start	0x54	0x52	0x02	Stop

Table 15. Up-line Response Table

Start bit	Address	Data high bytes	Data low bytes	CRC (first 2 bytes)	Stop bit
Start	0x55	0x00	0x01	0xB0	Stop

Note: The data is the result of restoring to factory values. The value is 1 if the restoration is successful and 0 if the restoration is not successful.

5) Read software version command, used to read the current software version, including downlink instructions and uplink data, as shown in Table 16 and Table 17.

Table 16. Downstream command list

Start bit	Address	Command high bytes	Command low bytes	Stop bit
Start	0x54	0xD1	0x00	Stop

Table 17. Up-line Table

Start bit	Address	Version high~low byte	Stop bit
Start	0x55	10 ASCII code	Stop

6) Read sensor number command, used to read the current sensor number, including downlink instructions and uplink data, as shown in Table 18 and Table 19.

Table 18. Downstream command list

Start bit	Address	Command high bytes	Command low bytes	Stop bit
Start	0x54	0xD2	0x01	Stop

Table 19. Up-line Table

Start bit	Address	SN high~low byte	Stop bit
Start	0x55	10 ASCII code	Stop

2.3.3 Actual code routine:

I^2C mode continuous read instructions:

Downlink command: 0x54 0x03 0x00 (0x54 is 0x2A move one left)

Uplink data: PPM3 PPM2 CRC PPM1 PPM0 CRC TEM1 TEM0 CRC (Every two bytes followed by CRC)

Concentration:

```
PPMCO2 = (uint)((((uint)PPM3)<<24) | (((uint)PPM2)<<16) |
         (((uint)PPM1)<<8) | ((uint)PPM0));
```

CRC check:

```
unsigned char Calc_CRC8(unsigned char *data, unsigned char Num)
{
    unsigned char bit, byte, crc=0xFF;

    for(byte=0; byte<Num; byte++)
    {
        crc^=(data[byte]);
        for(bit=8;bit>0;--bit)
        {
            if(crc&0x80) crc=(crc<<1)^0x31;
            else crc=(crc<<1);
        }
    }
    return crc;
}
```

2.4 UART communication protocol

2.4.1 Protocol Overview

The sensor supports UART communication, the communication baud rate is 1200, and the data format contains 8 data bits, no parity, and 1 stop bit. The protocol data are all in hexadecimal.

2.4.2 Protocol format

Table 20. UART protocol format

Frame head	Fixed code	Length	Command code	Data 1	……	Data n	Checksum (CS)
FE	A6	XX	XX	XX	……	XX	XX

Table 21. UART Protocol Format Description Table

NO.	Description	Define
1	Length	Data length
2	Command Number	Command
3	Data	Length of a read or write, with a variable length
4	Check Sum	Data accumulation and = fixed code + length + command number + data

2.4.3 Command list

Table 22. UART Command List

NO.	Function	Command
1	Read CO_2 concentration	0x01
2	Manual Calibration	0x03

3	Set the Calibration Mode	0x04
4	Factory Data Reset	0x05
5	Read the Software Version	0x1E
6	Read the Sensor Number	0x1F

1） Read CO2 concentration command, including sending and response, as shown in Table 23.

Table 23. UART reading concentration command table

Send	Reply	Description
FE A6 00 01 A7	FE A6 04 01 D1~D4 CS	CO_2 Concentration value=D1×256+D2； D3、D4 are retained

2） Manual calibration (single point calibration) command sets the reference point to a user-specified value, as shown in Table 24.

Table 24. UART manual calibration (single point calibration) command list

Send	Reply	Description
FE A6 02 03 D1 D2 CS	FE A6 00 03 A9	Single-point calibrated concentration value =D1×256+D2

3) Calibration mode setting command, including sending and responding, as shown in Table 25.

Table 25. UART calibration mode setting instructions

Send	Reply	Description
FE A6 02 04 00 D1 CS	FE A6 00 04 CS	D1=0 manual (single point) calibration; D1=1 automatic calibration

4) Restore factory settings command. When necessary, use this command to restore to factory values, as shown in Table 26.

Table 26. UART restore factory settings command

Send	Reply	Description
FE A6 00 05 AB	FE A6 00 05 AB	

5) Read software version instructions, including sending and responding, as shown in Table 27.

Table 27. UART read software version instructions

Send	Reply	Description
FE A6 00 1E C4	FE A6 0B 1E D1~D11 CS	D1~D10 is the ASCII code of the version number; D11 is reserved

6) Read the sensor number command, including sending and responding, as shown in Table 28.

Table 28. UART read number command

Send	Reply	Description
FE A6 00 1F C5	FE A6 0A 1F D1~D10 CS	D1~D10 is the ASCII code of the Serial number;

3. Dimension

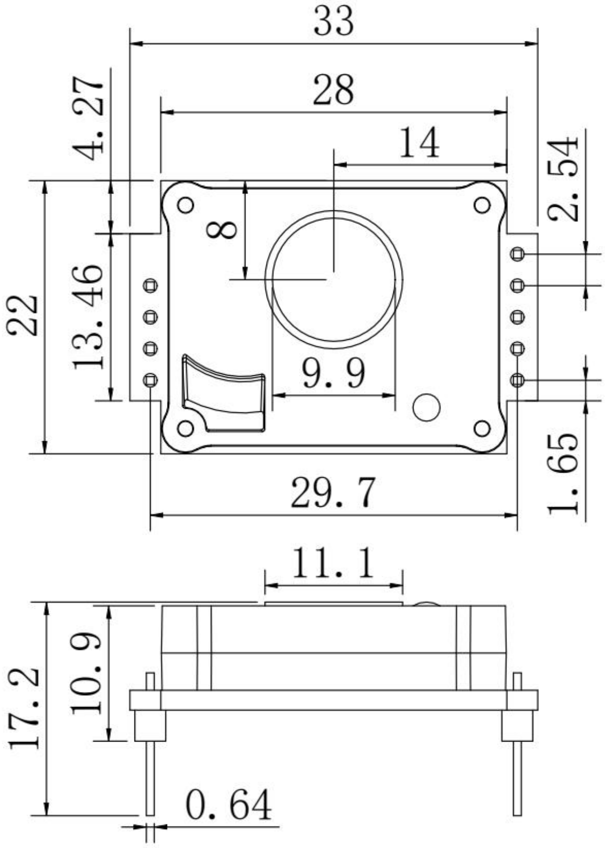

Figure 4. ACD1200 dimension(unit:mm； tolerance:±0.5mm)

4. Precautions

- Users must not disassemble the sensor to prevent irreversible damage.
- During manual calibration, the sensor must work continuously for more than 20 minutes in a stable gas environment with known CO_2 concentration.
- In manual calibration mode, it is recommended that the calibration cycle be no more than 6 months.
- The opening size of the air inlet and outlet of the equipment should not be smaller than the opening size of the sensor air inlet (shown as a circular window with a diameter of 9.9mm in Figure 4).
- The sensor should be kept away from heat sources and away from direct sunlight or other thermal radiation.
- Do not use the sensor for a long time in an environment with high dust density.
- It is prohibited to use wave soldering on the sensor. When soldering with a soldering iron, the temperature should be set below 350°C and the soldering time must be less than 3 seconds.
- It is recommended that customers use welding sockets to install sensors for maintenance.
- The data of the factory sensors have been tested and the data consistency is good. Please do not use third-party testing instruments or data as a comparison standard. If the user wants the measurement data to be consistent with the third-party testing equipment, the data can be calibrated based on the actual measurement results.
- Moisture-proof treatment has been taken on the sensor PCB. Silicone potting glue or acrylic conformal paint is used for coating. The thickness is not less than 0.15mm. Please pay attention to the protective coating during use.

Warning

Do not apply this product to safety protection devices or emergency stop equipment, and any other applications that may cause personal injury due to the product's failure. Do not use this product unless there is a special purpose or use authorization. Refer to the product data sheet and application guide before installing, handling, using, or maintaining the product. Failure to follow this recommendation may result in death and serious personal injury.

The Company will not bear all compensation for personal injury and death arising therefrom, and will exempt the company's managers and employees, affiliated agents, distributors, and other claims that may arise therefrom, including various costs, claims, lawyers' fees, etc.

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Description of Warranty Period of Main Components

Accessories Category	Shelf Life
ACD1200 IR CO_2 Sensor	12 Months

The company is only responsible for products that are defective when used in applications that meet the technical conditions of the product. The company does not make any guarantees or written statements about the application of its products in those special applications. At the same time, the company does not make any promises about the reliability of its products when applied to products or circuits.

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